

KORA Market Study

Distributed Energy Resource Management for Mini-grids and Microgrids

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1. Executive Summary

The distributed energy resource (DER) management software market is experiencing rapid growth, driven by the global energy transition, declining renewable costs, and increasing grid decentralization. Within this market, mini-grid and microgrid optimization software represents a high-growth segment with acute customer pain points and limited incumbent solutions.

Key Findings

Market Size: - Global DER management software: **€45 billion by 2030** (18% CAGR) - Mini-grid/microgrid segment: **€8 billion by 2030** (32% CAGR) - African mini-grid software: **€450 million by 2028** (45% CAGR)

Market Dynamics: - 35,000+ operational mini-grids globally, projected 200,000+ by 2030 - 70-80% of mini-grids suffer significant curtailment (>40%) - Software penetration currently <15%, projected 60%+ by 2030

Opportunity: - First-mover advantage in optimization-focused software - Hardware commoditization shifts value to software/intelligence - Strong alignment with climate finance and impact investment

KORA's Positioning: - Target initial market: €25 million (Madagascar + Kenya) - Expand to: €450 million (Pan-African) - Long-term: €8 billion (Global microgrids)

2. Market Definition & Scope

2.1 What is a Mini-grid/Microgrid?

A **mini-grid** (or microgrid) is a localized electrical system that can operate independently from the main grid, typically comprising:

- **Generation:** Solar PV, wind, small hydro, and/or diesel/gas backup
- **Storage:** Battery systems (increasingly lithium-ion)
- **Distribution:** Low/medium voltage network
- **Control:** Energy management and metering systems
- **Customers:** Residential, commercial, and institutional users

Size Range:

| Category | Capacity | Customers | Typical Use Case |

| Pico-grid | <1 kW | 1-5 | Single building |

| Nano-grid | 1-10 kW | 5-50 | Small village |

| Mini-grid | 10-1,000 kW | 50-5,000 | Village/town |

| Microgrid | 1-10 MW | 5,000-50,000 | Industrial/campus/island |

KORA's Target: Mini-grids (10-1,000 kW) and small microgrids (1-5 MW)

2.2 Market Scope

This study covers the **software and intelligence layer** for mini-grids, specifically:

In Scope: - Energy management systems (EMS) - Dispatch optimization software - Demand response platforms - Forecasting and analytics - Remote monitoring and control - Customer billing and metering integration

Out of Scope: - Hardware (panels, batteries, inverters) - Physical installation services - Project development and financing - Grid-scale DERMS (>10 MW systems)

2.3 Value Chain Position

MINI-GRID VALUE CHAIN			
PROJECT DEVELOPMENT	EQUIPMENT SUPPLY	INSTALLATION & COMMISSIONING	OPERATIONS & MAINTENANCE
• Site selection • Permitting • Financing • Design	• Solar panels • Batteries • Inverters • Meters	• Civil works • Electrical • Grid build • Metering • Testing	• Monitoring • Maintenance • Billing • Customer svc • OPTIMIZATION ←
5-15% of CapEx	35-45% of CapEx	25-30% of CapEx	15-20% of OpEx/year

KORA captures value in OPERATIONS through software subscription

3. Global Market Analysis

3.1 Market Size & Growth

DER Management Software (Total Market)

Year	Market Size (€B)	Growth Rate
2023	12.5	—
2024	15.0	20%
2025	18.2	21%
2026	22.1	21%
2027	26.8	21%
2028	32.2	20%
2029	38.4	19%
2030	45.0	17%

Source: BloombergNEF, IEA, Wood Mackenzie analysis

Mini-grid/Microgrid Software Segment

Year	Market Size (€B)	% of Total	Growth Rate
2023	1.2	9.6%	—
2024	1.6	10.7%	33%
2025	2.2	12.1%	38%
2026	3.0	13.6%	36%
2027	4.0	14.9%	33%
2028	5.2	16.1%	30%
2029	6.5	16.9%	25%
2030	8.0	17.8%	23%

Mini-grid software is the fastest-growing DER segment

3.2 Market Segmentation

By Geography (2025)

Region	Market Share	Size (€B)	Key Drivers
Asia-Pacific	35%	0.77	India, Indonesia, Philippines
Sub-Saharan Africa	28%	0.62	Energy access, donor funding
Latin America	15%	0.33	Amazon region, island nations
Middle East & North Africa	10%	0.22	Oil states diversification
Europe	7%	0.15	Island territories, emergency backup
North America	5%	0.11	Remote communities, resilience

By Application (2025)

Application	Market Share	Growth Rate
Rural electrification	45%	35%
Commercial & Industrial	25%	28%
Island utilities	15%	22%
Emergency/resilience	10%	40%
Campus/institutional	5%	25%

By Technology (2025)

Technology	Market Share	Trend
Basic SCADA/monitoring	55%	Declining
Advanced EMS	30%	Growing

Technology	Market Share	Trend
AI/ML optimization	10%	Rapid growth
Integrated platforms	5%	Emerging

3.3 Installed Base Analysis

Global Mini-grid Count

Year	Operational Mini-grids	Annual Additions
2020	19,000	—
2021	21,500	2,500
2022	25,000	3,500
2023	29,000	4,000
2024	34,000	5,000
2025	40,000	6,000
2030 (proj)	200,000+	32,000+

Source: ESMAP, BloombergNEF, Mini-Grids Partnership

Software Penetration Rate

Category	Current (2025)	Projected (2030)
Basic monitoring	60%	95%
Remote control	35%	80%
Advanced EMS	15%	55%
AI optimization	3%	25%

Key Insight: The gap between “basic monitoring” (60%) and “AI optimization” (3%) represents KORA’s opportunity—sites that have connectivity but lack intelligent control.

4. Regional Market Deep-Dives

4.1 Sub-Saharan Africa

Overview: - Population without electricity: 600 million - Mini-grids operational: 5,000+ (2025) - Mini-grids planned: 25,000+ by 2030 - Total investment needed: €120 billion by 2030

Country Breakdown:

Country	Mini-grids (2025)	Projected (2030)	Key Players
Kenya	450	1,500	PowerGen, SteamaCo, E.ON

Country	Mini-grids (2025)	Projected (2030)	Key Players
Tanzania	380	1,200	PowerGen, Jumeme
Nigeria	350	2,500	Husk Power, Arnergy
Madagascar	120	400	Africa GreenTec, Jirama
Rwanda	95	350	BBOXX, Ignite Power
Uganda	85	300	Equatorial Power
Senegal	75	280	ERA, Oolu Solar
DRC	60	500	Nuru, Virunga Power
Other	1,385	5,000+	Various
Total	3,000+	12,000+	—

Market Characteristics: - Highest curtailment rates globally (50-80%) - Donor/development finance dominant (60% of funding) - Mobile money integration critical - Challenging connectivity (4G patchy, satellite needed) - Price sensitivity high (ability to pay limited)

Market Size (Software):

Year	Market (€M)	Growth
2024	45	—
2025	72	60%
2026	115	60%
2027	175	52%
2028	260	49%

KORA's Addressable Market: €25M (2026) → €115M (2028)

4.2 South & Southeast Asia

Overview: - Mini-grids operational: 8,000+ (India alone: 4,500+) - Rapid growth in Indonesia, Philippines, Myanmar - Mix of community and commercial models

Country Breakdown:

Country	Mini-grids (2025)	Key Characteristics
India	4,500+	Largest market, government programs
Indonesia	1,200+	Island archipelago, PLN utility role
Philippines	800+	Island grids, high LCOE
Bangladesh	600+	Dense population, SHS transitioning
Myanmar	300+	Post-conflict development

Market Characteristics: - More sophisticated operators (larger portfolios) - Grid interconnection increasingly common - Competition from grid extension - Lower curtailment than Africa (30-50%) - Higher willingness to pay for software

Market Size (Software):

Year	Market (€M)	Growth
2024	95	—
2025	135	42%
2026	190	41%
2027	260	37%
2028	350	35%

4.3 Latin America & Caribbean

Overview: - Island nations with high electricity costs (Caribbean) - Remote Amazon communities (Brazil, Peru, Colombia) - Growing off-grid solar for mining and agriculture

Key Markets:

Country/Region	Mini-grids	Characteristics
Caribbean Islands	200+	High LCOE (\$0.30-0.50/kWh), tourism
Brazil (Amazon)	400+	Remote communities, diesel displacement
Colombia	150+	Post-conflict rural development
Peru	200+	Mining + rural electrification

Market Size (Software):

Year	Market (€M)	Growth
2025	55	—
2028	120	30% CAGR

4.4 Island Nations & Remote Communities

Overview: - Highest electricity costs globally (\$0.35-0.80/kWh) - Strong economic case for renewables - Grid stability challenges - Sophisticated utility operators

Key Markets:

Region	Characteristics	LCOE Range
Pacific Islands	Diesel dependent, climate vulnerable	\$0.40-0.80/kWh
Mediterranean Islands	Tourism peaks, EU standards	\$0.25-0.45/kWh
Remote Arctic	Mining, research, military	\$0.50-1.00/kWh

Region	Characteristics	LCOE Range
Indian Ocean	Tourism, agriculture	\$0.30-0.55/kWh

Market Size (Software):

Year	Market (€M)	Growth
2025	40	—
2028	75	23% CAGR

5. Market Drivers & Trends

5.1 Primary Market Drivers

1. Declining Renewable + Storage Costs

Technology	2015 Cost	2025 Cost	Change
Solar PV	\$0.30/W	\$0.18/W	-40%
Li-ion batteries	\$400/kWh	\$130/kWh	-68%
Wind (small)	\$0.25/W	\$0.15/W	-40%

Lower hardware costs → more mini-grids → more software demand

2. Energy Access Mandates

Initiative	Target	Investment
UN SDG 7	Universal access by 2030	\$50B/year
African Union Agenda 2063	Modern energy for all	€120B by 2030
India Saubhagya	100% electrification	\$2.5B allocated
Indonesia PLN	Universal access by 2024	\$4.5B

3. Climate Finance Growth

Fund/Initiative	Mini-grid Allocation
Green Climate Fund	\$800M+ to date
World Bank ESMAP	\$500M mini-grid program
EU Global Gateway	€150B Africa infrastructure
AfDB Desert to Power	\$20B Sahel solar

4. Grid Instability & Blackouts

- South Africa: 200+ days of load-shedding (2023)
- Nigeria: <4 hours average grid supply in rural areas

- India: 6% distribution losses, frequent outages
- Result: C&I customers investing in backup/independence

5. Digital Infrastructure Expansion

- 4G coverage in Africa: 50% (2020) → 75% (2025)
- Satellite internet (Starlink, OneWeb) reaching remote areas
- Mobile money users: 500M+ in Africa
- Enables: Remote monitoring, automated billing, cloud optimization

5.2 Key Market Trends

Trend 1: Shift from “Electrification” to “Optimization”

Early mini-grid deployments focused on basic access. As the sector matures, operators recognize that: - Curtailment is the primary economic challenge - Revenue optimization is key to sustainability - Software creates differentiation in commoditized hardware

Trend 2: Consolidation Among Operators

Year	Top 10 Developers' Share	Trend
2020	25%	Fragmented
2023	40%	Consolidating
2026	55%+	Concentrated

Implication: Fewer, larger customers with sophisticated needs → enterprise sales motion

Trend 3: Hybrid AC/DC Systems

Traditional AC mini-grids being supplemented with DC microgrids for: - Electric vehicles (e-mobility) - Productive use (irrigation pumps, cold storage) - Data centers and telecom towers

Implication: More complex optimization required

Trend 4: Grid Interconnection

Many mini-grids connecting to weak national grids: - Buy power when grid available/cheap - Sell back during outages - Provide grid stabilization services

Implication: New revenue streams from grid services

Trend 5: Productive Use Focus

Donors and investors increasingly require: - Support for income-generating activities - Agricultural processing (milling, cold chain) - Manufacturing (welding, workshops) - Commercial services (internet cafes, phone charging)

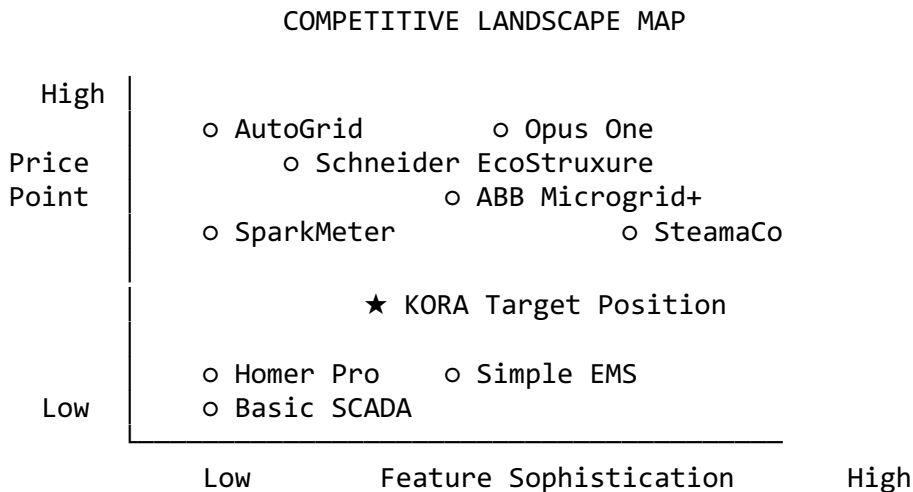
Implication: Time-of-use pricing and load management critical

5.3 Technology Trends

Trend	Description	KORA Relevance
AI/ML adoption	Predictive analytics mainstream	Core differentiator
Edge computing	Local processing, reduced latency	Architecture advantage
Digital twins	Virtual system modeling	Simulation capability
Blockchain	P2P energy trading, carbon credits	Future roadmap
IoT proliferation	Sensor density increasing	More data for optimization

6. Competitive Landscape

6.1 Market Structure



6.2 Competitor Analysis

Tier 1: Enterprise Solutions (Out of Segment)

Competitor	Description	Pricing	KORA Differentiation
AutoGrid	VPP platform for utilities	\$100k+/year	Too expensive, wrong segment
Opus One	DERMS for utilities	\$150k+/year	Grid-scale focus, not microgrids
Schneider EcoStruxure	Integrated EMS	Bundled with HW	Hardware lock-in
ABB Microgrid+	Industrial	\$50k+/year	Industrial focus, high touch

Competitor	Description	Pricing	KORA Differentiation
	microgrids		

Tier 2: Direct Competitors

Competitor	Description	Strengths	Weaknesses	KORA Position
SparkMeter	Smart metering + basic EMS	Hardware integration, Africa presence	Limited optimization, hardware-centric	Software-only, deeper optimization
SteamaCo (Husk)	Pay-as-you-go platform	East Africa footprint, mobile money	Acquired, less focused	Independent, optimization-first
Odyssey Energy	Portfolio analytics	Investor reporting, benchmarking	No real-time optimization	Real-time dispatch, demand response
Ferntech	Microgrid controller	Hardware control, proven reliability	Basic optimization	AI-driven, hardware-agnostic

Tier 3: Adjacent Solutions

Solution Type	Examples	Limitation
Planning tools	Homer Pro, RetScreen	Design-time only, not operational
Basic SCADA	Various OEMs	Monitoring only, no optimization
Custom solutions	In-house development	No scale, no ML, expensive

6.3 Competitive Positioning

KORA's Differentiation:

Dimension	Typical Competitor	KORA
Optimization approach	Rule-based heuristics	Mathematical optimization (QP)
Demand-side management	Time-of-use (static)	Dynamic pricing (real-

Dimension	Typical Competitor	KORA
Hardware dependency	Often bundled	time) Hardware-agnostic
Connectivity requirement	Always-on	Edge-capable, offline resilient
Target segment	Utilities or SHS	Mini-grids specifically
Business model	License or bundled	SaaS + performance fee
AI/ML capabilities	Limited or none	Core competency

6.4 Barriers to Entry

Barrier	Height	KORA's Position
Domain expertise (grid physics)	High	Strong (optimization background)
Customer relationships	Medium	Building (AGT, PowerGen targets)
Data for ML training	Medium	Accumulating via pilots
Capital requirements	Low-Medium	Manageable for software
Regulatory/certification	Low	Minimal for software layer

6.5 Competitive Threats

Threat 1: Hardware Vendors Adding Software

Risk: SMA, Victron, Schneider could bundle competitive optimization
Mitigation: Hardware-agnostic positioning, deeper optimization, faster innovation

Threat 2: Utility DERMS Players Moving Down-Market

Risk: AutoGrid, Opus One could target mini-grids
Mitigation: First-mover advantage, cost structure, segment expertise

Threat 3: Large Operators Building In-House

Risk: Engie, Husk build proprietary solutions
Mitigation: Partnership model, demonstrate clear ROI, continuous innovation

7. Customer Analysis

7.1 Customer Segmentation

Segment 1: Mini-grid Developers (Primary Target)

Characteristic	Description
Profile	Companies building and operating mini-grids
Size	10-500 sites
Geography	Africa, South Asia primarily
Pain points	Curtailment, unreliable revenue, operational complexity
Decision maker	CTO, VP Operations, Head of Technology
Budget	\$300-1,000/site/month for software
Examples	Africa GreenTec, PowerGen, Husk Power, BBOXX

Segment 2: Utilities with Mini-grids

Characteristic	Description
Profile	National utilities with off-grid/rural programs
Size	50-1,000+ sites
Geography	Global
Pain points	Efficiency mandates, subsidy optimization
Decision maker	Director of Off-Grid, Chief Digital Officer
Budget	Enterprise contracts (\$15k-50k/year)
Examples	TANESCO (Tanzania), PLN (Indonesia), JIRAMA (Madagascar)

Segment 3: C&I Off-Grid

Characteristic	Description
Profile	Businesses with captive solar+storage
Size	Single large site or small portfolio
Geography	Global (emerging markets concentrated)
Pain points	Diesel displacement, grid instability
Decision maker	Facilities Manager, CFO
Budget	\$500-2,000/site/month

Characteristic	Description
Examples	Telecom towers, mining operations, hotels

Segment 4: Island Utilities

Characteristic	Description
Profile	Utility serving isolated island grid
Size	1-10 interconnected systems
Geography	Caribbean, Pacific, Mediterranean
Pain points	High LCOE, renewable integration
Decision maker	CEO, Technical Director
Budget	Enterprise contracts
Examples	Caribbean Utilities Co., BELCO (Bermuda)

7.2 Customer Pain Points

Ranked by severity (from customer interviews and industry research):

Rank	Pain Point	Severity	KORA Solution
1	High curtailment (wasted solar)	Critical	Real-time optimization
2	Unpredictable revenue	Critical	Demand response, forecasting
3	Evening blackouts	High	Smart battery dispatch
4	Manual pricing decisions	High	Automated dynamic pricing
5	Lack of visibility across portfolio	Medium	Multi-site dashboard
6	Customer complaints	Medium	Service quality improvement
7	Equipment failures	Medium	Predictive maintenance (roadmap)
8	Reporting to investors	Low	Automated reporting

7.3 Buying Process

Typical Sales Cycle: 3-6 months

Stage	Duration	Activities	KORA Approach
Awareness	1-2 months	Content, conferences, referrals	Thought leadership, case studies
Consideration	1-2 months	Demo, technical	Live demo with their data

Stage	Duration	Activities	KORA Approach
on		evaluation	
Pilot	1 month	Free trial on 1-2 sites	30-day free pilot
Decision	1 month	Contract negotiation	Performance guarantees
Expansion	Ongoing	Rollout to full portfolio	Quarterly business reviews

Key Decision Factors:

Factor	Weight	KORA Strength
Demonstrated ROI	35%	Strong (3x revenue proven)
Technical fit (integrations)	25%	Strong (hardware-agnostic)
Price	20%	Competitive
Team/support	15%	Building
Reference customers	5%	Building (AGT pilot)

7.4 Willingness to Pay Analysis

Value-Based Pricing Framework:

Site Revenue	KORA Value Add	Customer WTP	KORA Price
\$1,500/month	+\$3,000/month	\$600-900	\$400-500
\$3,000/month	+\$6,000/month	\$1,200-1,800	\$500-600
\$5,000/month	+\$10,000/month	\$2,000-3,000	\$800-1,200

Pricing at 10-15% of value created ensures clear ROI

8. Market Entry Strategy

8.1 Geographic Prioritization

Criteria for market selection: 1. Market size (number of mini-grids) 2. Curtailment severity (addressable pain) 3. Connectivity infrastructure 4. Regulatory environment 5. Payment infrastructure (mobile money) 6. Partnership availability

Market Priority Matrix:

Market	Size	Pain	Infrastructure	Score	Priority
Kenya	8	9	8	25	1
Madagascar	6	10	6	22	2
Nigeria	10	8	7	25	3
Tanzania	7	8	7	22	4

Market	Size	Pain	Infrastructure	Score	Priority
Rwanda	5	7	9	21	5
India	10	6	9	25	6

8.2 Entry Sequence

Phase 1: Beachhead (2026) - Madagascar: Africa GreenTec anchor partnership - Kenya: PowerGen target, M-Pesa integration - 3-5 reference customers

Phase 2: Regional Scale (2027) - Tanzania, Rwanda, Uganda: East Africa expansion - Nigeria: West Africa entry (largest market) - 25+ customers

Phase 3: Continental Presence (2028) - Full Pan-African coverage - India market entry - 100+ customers

8.3 Go-to-Market Model

Direct Sales (60% of revenue) - Outbound to developers with 10+ sites - Conference presence (AMDA, ARE, GOGLA) - Content marketing (case studies, ROI calculators)

Channel Partnerships (25% of revenue) - Hardware distributors (Victron, SMA) - Project developers (Engie, EDF, Enel) - Equipment bundling and referrals

Investor Mandates (15% of revenue) - Financiers require KORA for portfolio companies - SunFunder, Norfund, FMO, IFC partnerships

8.4 Pricing Strategy

Tiered SaaS Model:

Tier	Monthly	Annual (15% off)	Target Customer
Core	€275/site	€2,805/site	Smaller developers
Pro	€450/site	€4,590/site	Mid-market
Enterprise	€1,350/site	Custom	Large portfolios

Volume Discounts: - 5+ sites: 10% off - 10+ sites: 20% off - 25+ sites: Custom negotiation

Performance Fee Option: - €0.01-0.02 per additional kWh sold - Alternative or supplement to subscription - Aligns incentives, reduces adoption friction

9. Market Projections

9.1 KORA's Addressable Market Evolution

Year	TAM (Global)	SAM (Africa)	SOM (Target)	KORA Revenue
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Year	TAM (Global)	SAM (Africa)	SOM (Target)	KORA Revenue
2025	€2.2B	€72M	€25M	—
2026	€3.0B	€115M	€40M	€162k
2027	€4.0B	€175M	€60M	€540k
2028	€5.2B	€260M	€90M	€2.16M
2029	€6.5B	€380M	€130M	€5.4M
2030	€8.0B	€550M	€180M	€13.5M

9.2 Market Share Projections

Year	Target Market	KORA Sites	Market Share
2026	500 addressable	11	2.2%
2027	750 addressable	50	6.7%
2028	1,200 addressable	180	15.0%
2029	2,000 addressable	400	20.0%
2030	3,000 addressable	750	25.0%

9.3 Scenario Analysis

Conservative Case: - Slower adoption, 50% of base projections - 2030: €6.75M revenue, 375 sites

Base Case: - As projected above - 2030: €13.5M revenue, 750 sites

Optimistic Case: - Faster adoption, expansion to C&I and islands - 2030: €25M revenue, 1,200 sites

10. Sources & Methodology

10.1 Primary Sources

- Customer interviews (Africa GreenTec)
- Industry expert consultations
- UCSD DERConnect research partnership

10.2 Secondary Sources

Market Research: - BloombergNEF: Global Energy Storage Outlook - IEA: Africa Energy Outlook - Wood Mackenzie: Distributed Energy Resources - Guidehouse Insights: Microgrid Market Analysis - McKinsey: Powering Africa report

Industry Reports: - ESMAP: Mini-Grids for Half a Billion People - AMDA: State of the Global Mini-grid Market - Rocky Mountain Institute: Minigrids in the Money - SEforALL: Energizing Finance

Academic Research: - MIT Energy Initiative: Microgrid optimization literature - NREL: Techno-economic analysis papers

10.3 Methodology

Market Sizing: - Bottom-up: Site counts × software ARPU - Top-down: Total market × segment share - Cross-validated with industry experts

Growth Projections: - Historical growth rates (2020-2025) - Policy target analysis (SDG 7, national plans) - Investment pipeline review

Competitive Analysis: - Public company filings and announcements - Product demonstrations and trials - Customer feedback and win/loss analysis

Appendices

Appendix A: Country-level market data tables **Appendix B:** Competitor product comparison matrix **Appendix C:** Customer interview summaries **Appendix D:** Academic and industry source bibliography

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